# IMC - Internal Monitoring Checklist

Review identification, purpose and classification guide

SIMULATED · v1.5

| | | | |
|---|---|---|---|
| **Study / protocol**<br>GLENOS-01 / GS-2026-01 | **CIP version**<br>CIP v4.2 | **Site / code**<br>Riverside Orthopedic Center / 014 | **IMC ID**<br>IMC-014-02 |
| **IMP ID / version applied**<br>IMP-014 v1.0 | **Review type**<br>☑ Scheduled ☐ Focused ☐ Triggered | **Review period / data cut-off**<br>12 Aug to 15 Oct 2026 / 15 Oct 2026 | **Date of review**<br>21 Oct 2026 |
| **Trigger or reason (if not scheduled)**<br>Not applicable: second quarterly review set in the IMP | | **Reviewer / role**<br>Dr Lena Ortiz / Sub-investigator (P-02) | **Controlled location**<br>Investigator site file, tab 3 (Quality) / IMC-014-02 |

## A Purpose and use

The IMC records one internal review of the site's own controls, run as the Internal Monitoring Plan (IMP) sets out. For each check in scope the reviewer rates three things separately: whether the control was done (execution), whether the records show it (evidence) and whether it worked (effectiveness). The IMC cites the IMP version it followed; it does not change the plan's rules after seeing the results.

It refers to records by ID and does not copy their content. It does not judge a single participant, image or event, which stay in their own records, and it does not replace sponsor monitoring, safety reporting or CAPA.

| Unit of analysis | Decision supported | Inputs | Outputs |
|---|---|---|---|
| One site, one review period or trigger, and the checks in scope. | Whether each control was done, evidenced and effective, and what must be corrected, sent on or verified next. | IMP, records named for each check, previous IMC and open actions. | Findings and actions (G), routes to the PI Oversight Template (POT), SRR, CAPA and PIR, and the effectiveness check (I). |

## B Classification guide

Rate the three dimensions separately for each check in E. A negative or uncertain rating always needs a finding, limitation or action.

| Rating | Use when | Minimum response |
|---|---|---|
| **Execution** | | |
| **Complete** | All planned activity for the period was done, even if it found a problem. | Rate evidence and effectiveness. |
| **Partial** | Part of the planned activity was done, or a smaller sample than the IMP without approval. | Record what is missing; action with owner and date. |
| **Not performed** | The activity was due and not done, or cannot be confirmed. | Action with owner and date; consider escalation. |
| **Not applicable** | The check does not apply to this phase or scope. | Reason in D. |
| **Evidence** | | |
| **Adequate** | Records confirm what was done and what was found. | None. |
| **Partial** | Records confirm only part of it, or are scattered or late. | Name the missing record; evidence action. |
| **Insufficient** | Records missing, contradictory or not retrievable. | Action; effectiveness is not assessable. |
| **Not applicable** | Only when execution is also not applicable. | Reason in D. |
| **Effectiveness** | | |
| **Effective** | The control works and no relevant signal is open. | None. |
| **Observation** | Minor, isolated point with no sign of control failure. | Record; action only if follow-up is needed. |
| **Not effective** | Recurrence, material gap, failed response or open signal. | Finding, action, owner, date and route. |
| **Not assessable** | Cannot be judged from the records available. | State the limitation; action to complete it. |

## B Classification guide (continued): combinations that change the response

| Execution · Evidence · Effectiveness | Meaning | Minimum response |
|---|---|---|
| **Complete · Adequate · Effective** | The control works and can be shown. | Record the result. |
| **Complete · Adequate · Not effective** | The control was done and recorded but did not prevent or catch the problem. | Finding and action; consider CAPA or SRR. |
| **Complete · Insufficient · Not assessable** | Said to be done, but the records cannot show it. | Evidence action; re-rate at the next review. |
| **Not performed · Adequate** | Only when records show that the activity did not happen. A record of absence never shows execution. | Action with owner and date. |

## C What each check looks for

A general guide. The site-specific criteria are in IMP section E and prevail over this table.

| # | Check | Execution means | Evidence may include | Effective when |
|---|---|---|---|---|
| 1 | **Participant entry: consent and eligibility** | Consent and eligibility of the participants included in the period were checked. | Signed consent forms, version log, eligibility checklist, PSEAR. | No procedure before consent; eligibility signed before surgery; any gap reported as a deviation. |
| 2 | **Delegation and training** | Study tasks done in the period were matched to the delegation log and training records. | Delegation log, training records, task records (ICDL, PFVR). | No task done before delegation and training; gaps closed and routed. |
| 3 | **Sterile release to use** | Release, package, expiry and storage checks for the period were reviewed. | ICDL, release record, storage and expiry record, substitution record. | Release status and any breach or substitution reconciled and routed. |
| 4 | **Implant traceability and component reconciliation** | Planned configuration, implanted components and accountability were compared. | ICDL, implant labels, accountability log, operative note, CRF. | Lot, UDI and configuration agree, or each discrepancy is explained and routed. |
| 5 | **Instrumentation and operating-room readiness** | Readiness, compatibility and malfunction records were reviewed. | ICDL, room readiness record, deficiency record. | Missing, incompatible or faulty equipment recorded and dealt with. |
| 6 | **Operative documentation** | Operative notes were read for configuration, decisions, substitutions and complications. | Operative note, ICDL, SART reference, CRF query. | Notes complete enough to read safety and outcome later. |
| 7 | **Imaging usability and timing** | Imaging timing, views and usability were reviewed. | IQL, imaging source, PFVR, repeat record. | Each unusable or late set has a disposition, repeat or limitation. |
| 8 | **Follow-up execution** | Follow-up outputs were reviewed for timing and components. | PFVR, FUCR, source and CRF. | Missing components have a cause and a recovery route. |
| 9 | **Endpoint data agreement** | A sample of endpoint-critical fields was compared with the source. | Source record, CRF entry, comparison sheet, query log. | Sampled fields agree, or each discrepancy is corrected and explained. |
| 10 | **Timeliness and closure** | Items due in the period were checked against their deadlines. | Query log, action log, CAPA log, time stamps. | Backlog visible, owned and inside the agreed limit. |
| 11 | **External context** | The planned search of external information was done. | Search record, field notice, sponsor letter, PI note. | Relevant information read against local data and sent on when needed. |
| 12 | **Complication monitoring** | Planned CSST reviews and their routing were checked. | SCML, CSST, SART. | Reviews on schedule; each signal has a route; no event left unindexed. |

## D Checks in this review and what was reviewed

Scope and sample follow the IMP; a different rule is also entered in IMP section I, never applied silently. Write how many units were reviewed out of how many, and the records and range used. A check left out needs the reason and who approved it.

| # | Check | In scope? | Sample reviewed (n of N) | Records, ID range and cut-off | If not in scope or sample changed: reason, approved by |
|---|---|---|---|---|---|
| 1 | Participant entry: consent and eligibility | ☑ Yes ☐ No | 3 of 3 | PSEAR-014-012, -019, -020; consent v3.0 | |
| 2 | Delegation and training | ☑ Yes ☐ No | 14 of 58 | Task list 12 Aug to 15 Oct; TRE-014-01 to -11 | Sample of 10 widened to all 4 tasks of S-03 (IMP expansion trigger). |
| 3 | Sterile release to use | ☑ Yes ☐ No | 4 of 4 | ICDL-014-005 to -008; storage log Aug to Oct | |
| 4 | Implant traceability and component reconciliation | ☑ Yes ☐ No | 16 of 16 | ICDL label sheets, accountability log, CRF implant pages | |
| 5 | Instrumentation and operating-room readiness | ☑ Yes ☐ No | 1 of 4 | ICDL-014-007 (every 5th by date) | |
| 6 | Operative documentation | ☑ Yes ☐ No | 2 of 4 | Latest note of S-01 and S-03 | |
| 7 | Imaging usability and timing | ☑ Yes ☐ No | 8 of 8 | IQL-014-021 to -028 | |
| 8 | Follow-up execution | ☑ Yes ☐ No | 6 of 11 | PFVR-014-014 to -024, random 6 | |
| 9 | Endpoint data agreement | ☑ Yes ☐ No | 5 visits, 118 fields | PFVR-014-015, -017, -019, -021, -023; CRF | |
| 10 | Timeliness and closure | ☑ Yes ☐ No | 10 queries, 1 SAE | Query report 15 Oct; SART-014-004 | |
| 11 | External context | ☐ Yes ☑ No | | | Deferred in IMP-014 v1.0: sponsor screens literature and vigilance (CIP §12.4). |
| 12 | Complication monitoring | ☑ Yes ☐ No | 1 of 1 | CSST-014-01 (30 Sep 2026); SCML snapshot 30 Sep | |
| 13 | Explant handling and return | ☐ Yes ☑ No | | | No explant in the period; check stays in the IMP. |
| 14 | | ☐ Yes ☐ No | | | |

Rows 13 and 14 are for checks the site added in the IMP. Use the same row number in E.

## E Review of each check

One row per check in scope. Leave rows out of scope blank. Cite record IDs; do not copy their content.

| # | Check and question | Execution | Evidence | Effectiveness | Finding, signal or limitation (record IDs) | Action ID · route · owner · due date |
|---|---|---|---|---|---|---|
| 1 | **Participant entry: consent and eligibility**<br>Was consent signed on the current version before any study procedure, and eligibility confirmed by the PI before surgery? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☑ Effective<br>☐ Observation<br>☐ Not effective<br>☐ Not assessable | 3 participants consented on v3.0 before any procedure; PI sign-off dated before surgery. | None |
| 2 | **Delegation and training**<br>Was everyone who did a study task delegated and trained for it on that date? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☐ Observation<br>☑ Not effective<br>☐ Not assessable | S-03 entered ICDL-014-007 on 2 Oct and 3 more tasks before release on 6 Oct (TRE-014-11). | A-02-01, A-02-02 · POT-014-07 · PI, coordinator · 31 Oct |
| 3 | **Sterile release to use**<br>Were package integrity, expiry, storage and any breach or replacement reviewed and reconciled? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☑ Effective<br>☐ Observation<br>☐ Not effective<br>☐ Not assessable | All 4 devices checked before opening; no expiry or storage breach. | None |
| 4 | **Implant traceability and component reconciliation**<br>Did implanted components match labels, lot or UDI, accountability and CRF, with discrepancies routed? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☑ Observation<br>☐ Not effective<br>☐ Not assessable | Glenosphere lot in CRF of ICDL-014-006 mistyped; label and log agree. Corrected at review (Q-014-31). | None; corrected |
| 5 | **Instrumentation and operating-room readiness**<br>Were instruments available, compatible and checked, and every substitution or malfunction recorded? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☐ Observation<br>☑ Not effective<br>☐ Not assessable | Loaner set for ICDL-014-007 had no trial liners; set check sheet does not list them. | A-02-03 · SRR R-09 · OR nurse lead · 4 Nov |
| 6 | **Operative documentation**<br>Does each operative note let a reader rebuild the final configuration and intraoperative decisions? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☐ Adequate<br>☑ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☑ Observation<br>☐ Not effective<br>☐ Not assessable | Note of S-03 (ICDL-014-007) does not state liner offset (Q-014-33). | Q-014-33 · Sub-I · 28 Oct |
| 7 | **Imaging usability and timing**<br>Were image sets taken in window, with the required views, and usable, with a disposition for any that were not? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☑ Observation<br>☐ Not effective<br>☐ Not assessable | IQL-014-026 taken 3 days late; usable, reason recorded. | None; watch at next review |

## E Review of each check (continued)

| # | Check and question | Execution | Evidence | Effectiveness | Finding, signal or limitation (record IDs) | Action ID · route · owner · due date |
|---|---|---|---|---|---|---|
| 8 | **Follow-up execution**<br>Were visits done in window with all endpoint-critical components, and attendance not taken as completeness? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☑ Observation<br>☐ Not effective<br>☐ Not assessable | PFVR-014-019: ASES missing at 6 weeks; recovery started. | A-02-04 · research nurse · 5 Nov |
| 9 | **Endpoint data agreement**<br>Did the sampled endpoint-critical fields agree between source and CRF, with discrepancies corrected and explained? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☐ Observation<br>☑ Not effective<br>☐ Not assessable | 2 Constant scores transcribed wrongly (PFVR-014-017, -021); both corrected. Expansion trigger met. | A-02-05, A-02-06 · CAPA-014-03 · coordinator · 31 Oct |
| 10 | **Timeliness and closure**<br>Were notes, data entry, queries, safety follow-up and actions closed on time, or escalated? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☑ Observation<br>☐ Not effective<br>☐ Not assessable | 4 of 10 queries open over 10 days; SAE reported in time. | Coordinator clears by 31 Oct |
| 11 | **External context**<br>Were safety notices, field actions and new literature on the device looked for and read against local findings? | ☐ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☐ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☐ Observation<br>☐ Not effective<br>☐ Not assessable | | |
| 12 | **Complication monitoring**<br>Did complication reviews take place as planned, and were their signals sent on? | ☑ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☑ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☑ Effective<br>☐ Observation<br>☐ Not effective<br>☐ Not assessable | CSST-014-01 held in the planned month; one signal sent with recipient ID. | None |
| 13 | Explant handling and return | ☐ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☐ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☐ Observation<br>☐ Not effective<br>☐ Not assessable | | |
| 14 | | ☐ Complete<br>☐ Partial<br>☐ Not performed<br>☐ Not applicable | ☐ Adequate<br>☐ Partial<br>☐ Insufficient<br>☐ Not applicable | ☐ Effective<br>☐ Observation<br>☐ Not effective<br>☐ Not assessable | | |

## F Signals across checks

Use only for a problem that does not fit in one row of E. Route: POT, SRR, CAPA, CSST, PIR, sponsor or safety route, or Not required with the reason.

| Signal | Seen? | What was seen · checks (#) | Immediate containment | Route · record ID · owner |
|---|---|---|---|---|
| **Same problem in more than one check** | ☐ Yes ☑ No | | | |
| **Participant safety concern** | ☐ Yes ☑ No | | | |
| **Endpoint or data reliability concern** | ☑ Yes ☐ No | Transcription errors and a missing ASES (8, 9) | All entries since Aug second-checked | PIR, CAPA-014-03 · coordinator |
| **Device traceability or deficiency concern** | ☐ Yes ☑ No | Lot error corrected at review (4) | | |
| **Need for closer PI oversight** | ☑ Yes ☐ No | Tasks done before release (2) | S-03 tasks paused until released | POT-014-07 · PI |
| **Need for CAPA or a change in the SRR** | ☑ Yes ☐ No | Trial liners missing (5); transcription (9) | Trial liners added to the set by hand | SRR R-09 update; CAPA-014-03 · coordinator |

## G Action log

Every action has an owner and a due date. CAPA content stays in the CAPA log; write its ID here.

| Action ID | From (# / F) | Action | Type | Owner (role) | Due date | Linked record | Status |
|---|---|---|---|---|---|---|---|
| A-02-01 | 2 | Review the 4 tasks done by S-03 before release, with the PI | ☑ Containment ☐ Corrective ☐ Preventive ☐ Verification | PI | 22 Oct 2026 | POT-014-07 | ☑ Open ☐ Closed ☐ Transferred |
| A-02-02 | 2 | Check the TRE release before a name goes on the task roster | ☐ Containment ☑ Corrective ☐ Preventive ☐ Verification | Coordinator | 31 Oct 2026 | Delegation log rev. 8 | ☑ Open ☐ Closed ☐ Transferred |
| A-02-03 | 5 | Add trial liners to the loaner-set check sheet | ☐ Containment ☐ Corrective ☑ Preventive ☐ Verification | OR nurse lead | 4 Nov 2026 | SRR R-09 | ☑ Open ☐ Closed ☐ Transferred |
| A-02-04 | 8 | Recover ASES for PFVR-014-019 or record the limitation | ☐ Containment ☑ Corrective ☐ Preventive ☐ Verification | Research nurse | 5 Nov 2026 | PFVR-014-019 | ☑ Open ☐ Closed ☐ Transferred |
| A-02-05 | 9 / F | Second check of every endpoint entry since August | ☐ Containment ☑ Corrective ☐ Preventive ☐ Verification | Coordinator | 31 Oct 2026 | CAPA-014-03 | ☐ Open ☐ Closed ☑ Transferred |
| A-02-06 | 9 | Sample 5 more visits for endpoint agreement | ☐ Containment ☐ Corrective ☐ Preventive ☑ Verification | Sub-I | 15 Jan 2027 | IMC-014-03 | ☑ Open ☐ Closed ☐ Transferred |

# IMC - Internal Monitoring Checklist

Outcome and certification

SIMULATED · v1.5

## H Review outcome

The outcome comes only from the checks in scope.

| | |
|---|---|
| **Overall outcome** | ☐ Controls effective ☐ Observations only ☐ Action required ☑ Escalation required |
| **Checks needing follow-up (#)** | 2, 5, 6, 8, 9, 10 |
| **Summary of findings** | Most controls worked. Two need correction: a new surgeon did study tasks before release, and the loaner set check does not cover trial liners. Endpoint transcription failed twice in 118 fields and goes to CAPA. No participant safety concern. |
| **PI review required?** | ☑ Yes, POT ID: POT-014-07 ☐ No, reason: |
| **Next IMC due** | 15 Jan 2027 |

## I Effectiveness check of actions

An action is closed only when later evidence shows it worked, not when it is done.

| Action ID | Check date | Evidence reviewed (record ID) | Result | Reviewer |
|---|---|---|---|---|
| A-01-03 | 30 Sep 2026 | IQL-014-021 to -028: no repeat needed | ☑ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable | L. Ortiz |
| A-02-01 | 29 Oct 2026 | POT-014-07; TRE-014-11; S-03 tasks after 6 Oct | ☑ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable | L. Ortiz |
| | | | ☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable | |

## J Reviewer certification

I confirm that this review covered the stated scope and period, that the records cited were reviewed in proportion to the checks in scope, and that every negative or uncertain rating has a finding, limitation, action or route.

☑ I did not do the work I reviewed. ☐ Exception recorded in POT:

**Reviewer**

Name / role: Dr Lena Ortiz / Sub-investigator

Signature / date:

*L. Ortiz* 21 Oct 2026

**Principal Investigator (if required)**

Name: Dr Alex Morgan

Signature / date:

*A. Morgan* 22 Oct 2026

File this checklist with the records it cites. A focused or triggered review does not replace the next scheduled IMC unless H changes its date.